

# Review of software tools and methods used in operational support systems developments

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*This paper provides some background to the diversity of software development environments and tools used within BT. Past and present approaches to the subject are described, along with some of the lessons learned along the way. The paper explains why BT is concerned to use modern tools to give higher productivity and describes the work going on to standardise the operational support systems infrastructure and reduce the large number and variety of tools used. The company's huge software portfolio, covering systems of all sizes from standard (but very large) invoicing systems to complex systems which interact in real time with the voice and data networks, means that BT has to live with a diverse set of software environments and the large variety of tools which go with them. The challenge is to control the diversity in a rapidly changing industry.*

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## 1. Introduction

This paper sets out to share past and present approaches to the use of software tools and methodologies within BT, together with experience along the way. BT is a large company which has software engineering at its core. The size and variety of BT's software portfolio makes it impossible to choose one methodology and one toolset for all software engineers. The company has to live with and manage diversity.

## 2. Background and context

There are some 6000 software engineers in BT handling a software portfolio of around 1300 systems. There are about 500 operational support systems (OSS) in the portfolio covering standard business systems like order handling and invoicing through to the more telecommunications-specific systems which interface to and interact with the network. These systems cover batch processing, on-line transaction processing and real-time systems. BT needs methods and tools to handle strategic systems to get products to market quickly, methods and tools which support innovation and prototyping of new services like video on demand, methods and tools which will produce robust and reliable software for key operational systems like network traffic management, pricing and invoicing, as well as the support systems like payroll and personnel.

Twenty five years ago it was simple. There was the waterfall method and the only software tools available from vendors were language compilers, link-editors and sorts. Today, every consultancy firm can offer its own method for process re-engineering [1] and the consequential systems development. The industry has information engineering

methods from Martin and Texas Instruments, the iterative method of Boehm [2], real-time methods like Yourdon [3], object-oriented methods from Jacobsen, Booch [4], Rumbaugh and Martin. There is a UK method for rapid application development called the dynamic systems development method (DSDM) which is rapidly being accepted internationally [5]. For the last ten years or more vendors have marketed tools to service each of these methods and some have tools which serve one part of the life cycle only. There are software engineering work benches, maintenance work benches, testing tools, CASE tools, integrated CASE tools, etc. It is difficult to know how to choose when the IT industry and the market for all these tools is moving rapidly, and whatever toolset you pick today will be overtaken tomorrow.

BT has been using software engineering in its OSS since the mid-1950s, and some of its systems are over ten years old, with one of the support systems having reached 20 years old. Thus BT has a legacy of systems built with a small diversity of methods and a much larger diversity of software tools. At this stage it is just about manageable. However, the diversity of tools available is now so great and the changes in the software industry so rapid that BT is having to put new controls on to the take-up of tools and methods.

## 3. Why use methods and tools?

Industry benchmarking has shown that the productivity levels shown at Fig 1 can typically be achieved using the various generations of languages and tools. It can be seen that it is

clearly within BT's interest to use more sophisticated tools and automate the software development process. However, there is no point in increasing productivity in one part of the life cycle if the time saved is only wasted later trying to learn how to use and integrate the large variety of tools needed.

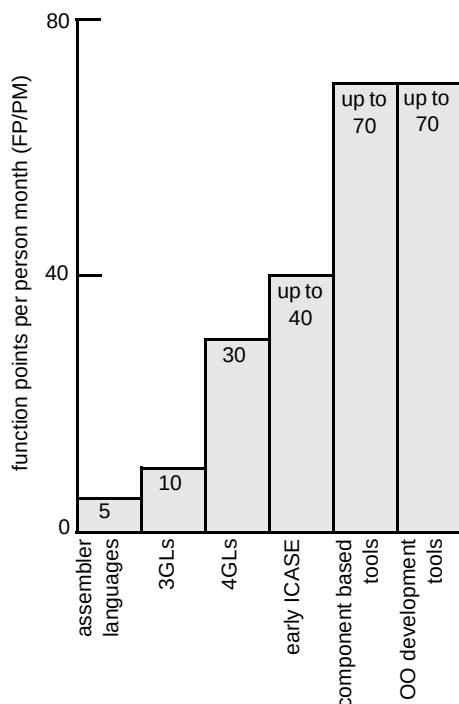


Fig 1 Typical productivity levels attained from current languages and tools .

Benchmarking activity has shown that world-class productivity levels (about 70 FP/PM) have been reached by development units of about 200 people using one rapid development method and one integrated toolset which covers the full life cycle. As the developers build up experience with the method and toolkit, their productivity increases. It does not seem to matter which integrated tool or rapid method is used as long as staff are fully trained and use the tool consistently for a long enough period to move up the experience curve.

#### 4. What is meant by tools?

Software tools cover a wide range. Choosing them and integrating them is not simple. Both the development and operational environments have to be considered. Within each of these there are software tools which have to be mixed and matched depending on the hardware and software environment being used and the methodology being adopted. BT's prime environments are IBM's MVS with either IDMS or DB2 as a database on the mainframe, the open systems management framework (OSMF) Unix with Oracle as the database for medium-sized or mid-tier systems and Microsoft's products on PCs on the desktop. Digital's VMS is used for many of the network systems.

In today's distributed environments hooks have to be put in to use system management software and the large and complex issues of configuration management from requirements through to version control of software releases also has to be considered. BT needs graphic user interface (GUI) builders, terminal emulators, design tools and code generators. Testing tools are also required like code scanners, static analysers and test harnesses. Finally, with the large and complex systems BT has to develop and support, it needs tools which assist with systems integration across the three tiers. The more of the above that can be covered using one tool the better.

#### 5. Major methods and tools BT have tried

In the late 1980s BT developed its own standards for the waterfall methodology, called TELSTAR. Almost as soon as it was written TELSTAR needed updating with new techniques like prototyping and its maintenance became a burden. New updates of TELSTAR stopped in 1990. It is still on many people's bookshelves and a hypertext version is available. It is still consulted and referred to in quality plans today and contains much good practice for developers using the waterfall methodology. An attempt to automate TELSTAR using a proprietary ICASE failed at the time, so limited automation was achieved using life cycle specific tools like Excelerator, APS and Bachmann.

At the same time information engineering and integrated CASE were being hailed as some new silver bullets and a couple of large systems were developed using TI's IEF. Even in the late 1980s the waterfall model was not appropriate for real-time systems which interface with the telephone network. The network engineers adopted Yourdon's method and their most favoured development tool was Software through Pictures. None of the above tools helped much with testing or configuration management and a host of other software tools were tried alongside these major toolkits. Table 1 shows the major methods and tools used over the last ten years. It is not an exhaustive list.

Table 1 Methods and tools used in BT over the last ten years.

| Method                                                          | Tools                                                                          |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------|
| waterfall (TELSTAR)                                             | Intersolv's Excelerator and APS. Bachmann's Maestro. Oracle CASE. Oracle forms |
| information engineering                                         | TI's IEF. Oracle Designer 2000                                                 |
| Yourdon (real-time systems)                                     | Software through Pictures                                                      |
| object-oriented analysis and design                             | Rational Rose                                                                  |
| dynamic systems development method (DSDM) for rapid development | TI's Composer. Oracle Developer 2000                                           |

## 6. Integrating and using tools

The following case is an example from the OSS billing area, where BT has successfully used and integrated tools across several hardware platforms and achieved high productivity levels in spite of the complexity of the environment.

### 6.1 Billing for interconnect traffic

Deregulation of the telecommunications industry within the United Kingdom has resulted in BT and other licensed operators (OLOs) passing large volumes of traffic between their networks. While this interconnection is transparent to the customer, costs need to be recovered by the parties involved.

BT gains control of interconnect revenues and payments using the inter-network call accounting (INCA) system. INCA also provides valuable management information which is already accessible to support interconnect contracts and commercial decisions. This billing system was developed by BT and has been independently audited to BSI ISO9001.

INCA consists of a series of generic processes, procedures and systems that provide accurate, timely and auditable billing information on wholesale interconnect telephony services. The main challenge has been to design a highly scalable system, capable of coping with the huge volume of data every day and rapid growth of the market, that is flexible enough to cater for a fast-changing and evolving business area. The final architecture of the system is a complex and innovative integrated solution.

The application software is a major CASE (computer aided software engineering) development utilising Texas Instruments' IEF/Composer tool combined with an advanced GUI-based report-generation package, Business Objects [6]. Some of the later on-line transactions utilise advanced three-tier client/server techniques based around TI's Arranger product. Itemised records are then held on optical disk to back up the summary data in case of inquiry or dispute. The optical disk system has capacity to store four billion individual call records. Record level retrieval is then made possible through a highly customised version of Dorotech's Dorokey software. INCA has been implemented over the last four years in a phased manner to meet the fast-evolving business area and to exploit the leading-edge technology. BT and the INCA systems are now regarded as world leaders in their field. The suppliers, Hewlett Packard, Texas Instruments and Oracle, also use the INCA system as a world-wide reference site to demonstrate the use of their technologies on a high-volume, strategic, commercial system.

This system is complex, and, though the development environment is four years old, it continues to deliver some of BT's highest productivity metrics, in terms of delivered function points per person month.

## 7. Outlook for the future

BT has to live with diversity, but, unless some attempt is made to minimise that diversity, the company could face problems as the software industry adopts new methods and tools and others become obsolete. The latest interest is in object orientation as a method, and many vendors have tools which support it. The Internet is another area for tools explosion as more and more software is developed for it. Transition-engineering products are also, of course, being looked at as a way of slowly rebuilding the essential components of BT's OSS legacy systems, as and when feasible.

For OSS, diversity is being controlled by putting an infrastructure baseline together which the majority of developers must use. This will form the OSS software infrastructure for the company. Those involved in R&D are allowed to roam outside that infrastructure in the cause of their research and new product development, but any changes that are needed to the baseline as a result must be controlled. The infrastructure covers all three layers of the three-tier client/server model as well as the networks which link the distributed components.

Having learned from the TELSTAR experience that it is too expensive to develop and maintain an in-house method, industry methods are now being adopted by OSS and vendors are being influenced in both the method and tools. For example, BT have joined, and now hold the chair of, the Dynamic Systems Development Consortium (DSDM) which has developed a UK method for rapid application development and this consortium is working with tool vendors to support the method. Tool vendors like TI and Oracle are in the consortium and have undertaken to provide automated support for DSDM.

## 8. Conclusion

BT is similar to many other large international companies who have been using software within their operations since the 1950s. The company depends on software. BT has a constant need to enhance existing software and develop new software and is working on platform architectures to give more flexibility. Fast tools and methods are needed to deliver the software in as short a time as feasible. There will never be a complete toolset. Time and the industry will have moved on.

## References

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## OSS DEVELOPMENT SOFTWARE TOOLS AND METHODS

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Sue Valiant has had a varied career in BT both managing software development and managing some of the support organisations like Applications and Data Architectures and Methods and Tools. One of her current roles is in managing the Tools and Development environment domain of BT's Infrastructure Programme. She also leads for Systems Engineering in benchmarking and metrics for IT Development.

She is a member of the British Computer Society, a Fellow of the Royal Society of Arts, and has a BA in History and an MBA.